

Algorithms for Sonifying Objects into Spatial Audio using Head- Related Transfer Functions

Algoritmer för att Sonifiera Objekt till Spatiellt Ljud genom Huvud Relaterade Överföringsfunktioner

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Abstract

This thesis investigates algorithms for sonifying objects into spatial audio. The primary focus has been to examine how Head-Related Transfer Functions (HRTFs) could be interpolated while maintaining accurate spatial localization cues. More specifically, how different interpolation methods compare when interpolating spatial localization cues. The findings from the experiments showed that using a neural network to predict the spectrum of the HRTFs provide high spectral accuracy and is superior in capturing the localization cues.

Furthermore, because human hearing cues are highly individual and depend on anatomical features such as ear and head shape, this thesis explores how a personalized neural network can improve HRTF interpolation. Utilizing the HUTUBS database, a model was trained to predict the individual spectrum of the HRTF. The results of the thesis showed that the model improved the shape of the magnitude interpolation. However, the timing differences between filters were inaccurate.

This thesis also investigates how the input layer of the anatomical features could be optimized to both increase prediction accuracy, but also to ease measurements for data collection.

Overall, the findings demonstrate the potential of the proposed personalized model. However, further work is required to improve phase and timing prediction, as well as to expand the training dataset.

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1 Introduction

A snapping twig, a passing car, or a familiar voice in a crowded room. The remarkable auditory system of humans helps us to precisely localize the direction these sound sources. Through different acoustical cues, the auditory system interprets sound sources and provides a foundation for human spatial awareness. These cues can be measured through *Head Related Transfer Functions* (HRTFs), yielding a set of transfer functions that can filter mono sound to directional stereo sound.

This thesis investigates different interpolation methods for HRTFs and also how a personalized neural network can improve localization cues and spectral accuracy.

1.1 Motivation

Hearing is one of human-kind's most sensitive senses. It has made humans highly efficient at interpreting auditory cues, resulting in improved spatial awareness. This has been vital in protecting humans from threats throughout history. In the early 20th century, Lord Rayleigh laid the foundation for the understanding of how our auditory system interprets spatial information [1]. Lord Rayleigh established what we today call *Interaural Time Difference* (ITD) and *Interaural Level Difference* (ILD) which are the cues used for horizontal sound localization [2]. This was later expanded to the lateral plane [3] and today the HRTF is widely used to determine the transfer function from a given source to the ear.

This thesis was carried out in cooperation with fellow master's student Edvard Nilsson [4]. Together with Nilsson, a system is developed for a *Raspberry Pi 5* [5] fitted with *Rpi HIFI DAC Hat* [6] and the *Raspberry Pi AI Camera* [7], see figure 1.1. Using the camera, Edvard developed an algorithm for object detection in images. From the detected location, the direction was extracted. These directions are then interpreted by the system in this thesis and produces two digital filters, one filter for each ear canal. These filters together add spatial localization cues creating a sense that a sound is played from the direction.

The aim for this type of project is to build a foundation for a system that can translate visual localization to audio localization. A foundation that later can be used in systems helping visually impaired humans, primarily in controlled environments, but potentially everyday life.

In paralympic cross country skiing, visually impaired athletes compete with a guide skier. The guide skier's purpose is to guide the visually impaired athletes through the course. As

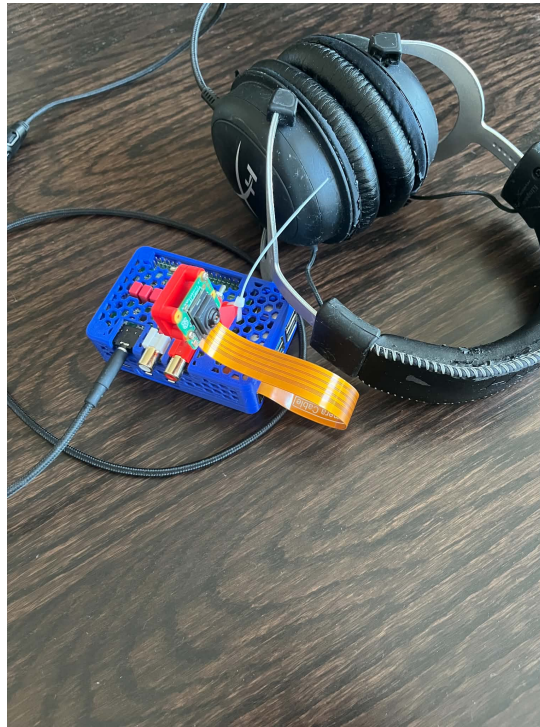


Figure 1.1: Image of the hardware used in this thesis

the course has course markers on the edge, a system utilizing the algorithms developed in this thesis and Nilssons thesis[4], can potentially replace the guide skier by describing the track by different sounds.

1.2 Aim

The aim of this thesis is to create a set of algorithms that utilizes the HRTF to add spatial cues by filtering mono sound to directional stereo sound. These algorithms are then used in a system[4] to sonify objects, enabling interpretation of surroundings using hearing instead of vision. In the report, different approaches for sonifying objects are compared to give a baseline for further development.

1.3 Research questions

This thesis aims to answer the following questions:

1. How can HRTF points be interpolated meanwhile spectral accuracy in terms of localization cues be maintained?
2. How can a personalized HRTF model improve spectral accuracy for sound localization compared to a generalized HRTF?
3. How can the number of anthropometrics in a personalized HRTF model be minimized while maintaining localization accuracy?

1.4 Delimitations

This thesis has several delimitations. First, the study relies on existing HRTF datasets rather than individual HRTF measurements. As a result, measuring individual HRTF measurements will not be conducted due to time and resource constraints.

The second delimitation is that environmental cues such as reflections and different sound transmission mediums will not be considered in this thesis. Instead, the focus is on accurately modelling hearing cues based on personal features and direction.

The third delimitation is that practical experiments on people will not be conducted. Even though it would be a very interesting experiment and would add to the report, time is a limiting factor.

2 Theory

This chapter provides the theoretical foundation of this work. It covers necessary definitions and principles to understand the project and the applied work.

2.1 Sonification

In this work, sonification is a vital concept. Kramer et al. define sonification as “the use of nonspeech audio to convey information” [8]. This definition is central to the project, as the desire is to convey information about an object or action to the user through sound. More specifically, sonification in this project will be used to communicate spatial localization cues. These localization cues are then processed through digital filters to create a spatial sound representation.

2.2 Spatial Hearing

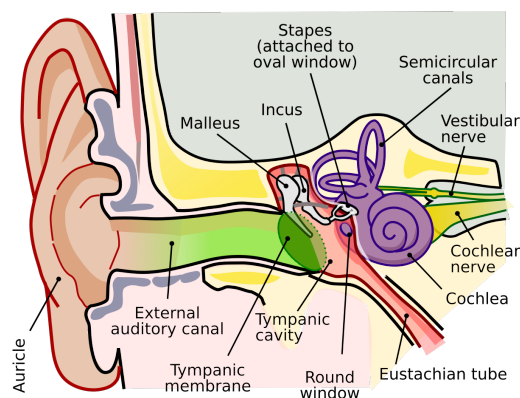


Figure 2.1: Visualization of the human auditory system. (Source: Chittka & Brockmann, PLoS Biology[9], CC BY)

Spatial hearing refers to the ability to perceive and localize a sound source.

The translation is done by different elements of the ear. When sound arrives at the ear, it is collected by the outer ear (auricle/pinna and external auditory canal). The sound waves travel through the external auditory canal and hit the tympanic membrane, causing vibrations in the middle ear. These vibrations are transmitted through the ossicles (malleus, incus, and stapes) in the middle ear. The vibrations cause fluid waves of the cochlea located in the inner ear. Based on the frequency of the vibration, different parts of the cochlea are affected. These fluid waves cause hair cells to bend in the cochlea, which converts the energy to electrical signals. These signals travel through the auditory/cochlear nerve to the brain, which interprets the signals as sound [10].

2.3 Localization cues

When sound reaches the auditory system, the brain extracts information from both ears to determine the direction of the sound source. These cues are divided into two different categories: binaural cues and monaural spectral cues.

Binaural cues mainly consist of *Interaural Time Difference* (ITD) and *Interaural Level Difference*. The auditory system correlates the information of the two ears and maps the difference to a direction.

Monaural spectral cues are processed in one ear independently of the other. Monaural cues convey information used mainly for vertical and antero-posterior sound perception [2], but also to determine distance [11].

2.3.1 Interaural Time Difference

Interaural Time Difference (ITD) refers to the time difference of arrival of a sound at the two ears. ITD is the dominant cue used for horizontal sound localization[12]. Localization based on ITD has been shown to be mainly dominated by low frequencies at approximately 1.5 – 2 kHz [11].

Physically, ITD occurs because sound will reach the closest ear before the other ear. Simplifying the head as a rigid sphere with radius r_h , Woodworth derived an expression for the ITD τ_{ITD} as a function of the azimuth θ and speed of sound c in equation 2.1 [13].

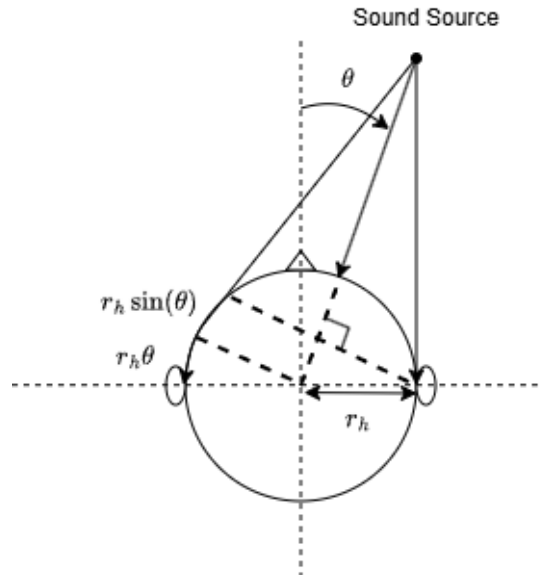


Figure 2.2: Visualization of Woodworth's formula, where r_h is the head radius and θ is the azimuth angle.

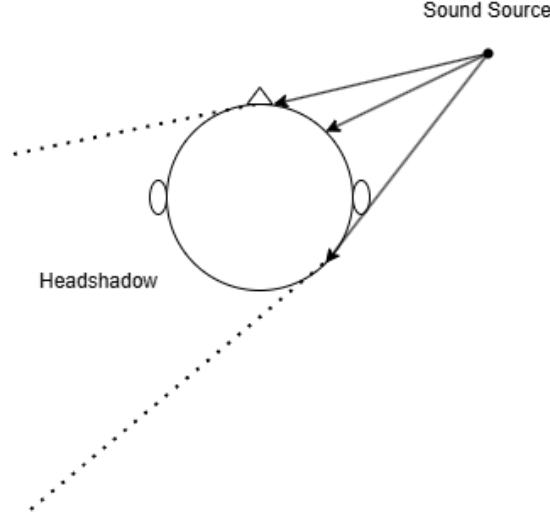


Figure 2.3: Visualization of the head shadow effect

$$\tau_{ITD} = \frac{r_h (\theta + \sin \theta)}{c} \quad [\text{s}] \quad (2.1)$$

Woodworth's formula was derived from making the assumption that the head is a rigid sphere, see figure 2.2

It is important to understand that Woodworth's model assumes the human head is a rigid sphere, which practically is inaccurate. The human head is neither round nor symmetrical. When inspecting the head from a top view, the shape can more be described as an ellipsoid. Furthermore, each individual person has a differently shaped head creating an individual formula for each person. It is also important to understand that Woodworth's formula only models the azimuth plane. For a correct physical model, the elevation φ has to be included to create an accurate representation.

2.3.2 Interaural Level Difference

Interaural level difference (ILD) refers to the pressure difference of the sound collected at each ear according to equation 2.2. The difference arises from the phenomenon called the *head shadow effect*. The head shadow effect arises as the head creates an acoustic shadow effect for the ears, see figure 2.3. This effect attenuates the signal, primarily at higher frequencies as shorter wavelengths is affected more by the head shadow effect than longer wavelengths. [14]

$$\text{ILD}(H_L(f), H_R(f)) = 20 \log_{10} \left(\frac{\|H_L(f)\|_2}{\|H_R(f)\|_2} \right) \quad [\text{dB}] \quad (2.2)$$

As a result of the head shadow effect, the perceived sound intensity is greater for the ear localized closest to sound source.

2.3.3 Elevation and Antero-Posterior cues

When interpreting sounds using ITD and ILD, the auditory system can accurately determine horizontal positions. However, it can not specify elevation and antero-posterior localization as many spatial positions produce identical interaural differences. Accurate elevation and antero-posterior localization are achieved through spectral cues. These spectral cues are

created by directional filtering of sound by the pinna through reflections creating peaks and notches in the magnitude. Each peak is interpreted as a direction and elevation by the human brain. [15]

2.3.4 Distance cues

Distance perception is built on three main cues. The first is sound intensity. As sound travels, the sound attenuates due to geometrical spreading. The intensity I of an arriving sound can be calculated with the power of the sound source P and the distance from the source r according to equation 2.3 [16], leading to an intensity decrease of approximately 6 dB for double the distance [15].

$$I = \frac{P}{4\pi r^2} \quad (2.3)$$

The second cue is spectral attenuation. As sound travels, higher frequencies are known to attenuate more than lower frequencies [17].

The third cue is reverberation. Sound takes different paths through reflections of surfaces before arriving at the ear. When arriving at the ear, these reverberated sound waves are time delayed and attenuated due to the reflections. Reverberation will not be included in this work.

2.4 Head-Related Transfer Function

The Head-Related Transfer Function (HRTF), or its time-domain counterpart the Head-Related Impulse Response (HRIR), is a digital filter describing how incoming sound from a spherical point $\mathbf{p} = (r, \theta, \phi)$ is filtered by the anatomical features of the upper body, head, and pinna before reaching the ear canals of a subject. A collection of sampled HRTFs are referred to as a HRTF dataset of a subject. A HRTF dataset is stored in the standardized *Spatially Oriented Format for Acoustics* (.sofa) file format [18].

The HRTF dataset collection for a subject can be done in several ways. In this work, HRTF dataset collection were not performed. Instead, publicly available HRTF datasets through the HUTUBS (Huawei, Technische University of Berlin, and Sennheiser) database[19][20] were used. The HRTF datasets in the HUTUBS database were measured in a controlled environment using a speaker-based setup. Each subject was seated in a fixed position with in-ear microphones in each ear. The speakers emitted sound that the microphones captured and the HRTF for each speaker position was calculated. The subject independent HRIR $h_{\text{HRIR}}(r, \theta, \phi)$ was saved in a .sofa file with all positions of the speakers.

2.5 Neural Networks

In this section, different terms related to neural networks are defined. Here, the different neural network architectures are described. For all networks in this work, the ADAM optimizer [21] is used for training of all models. In all networks, the chosen activation function between layers is the Rectified Linear Unit (ReLU) [22]. All networks were developed using pytorch [23].

2.5.1 Multilayer Perceptron

A multilayer perceptron (MLP) is a neural network structure in which all nodes in each layer are fully connected to neighbouring layers. The MLP consists of an input layer, followed by a number of hidden layers, and ends with an output layer. Each layer consists of a number of nodes. As each layer is fully connected, each node in the layer is a weighted sum of each node in the prior layer followed by a non-linearity. [24]

2.5.2 Convolutional Neural Network

A convolutional neural network (CNN) is a feedforward network architecture that utilizes kernels with learnable weights for prediction [25]. As a result of using kernels, the number of learnable weights are significantly decreased based on number of kernels and kernel size. Each CNN layer consists of N number of kernels. Each kernel produces a corresponding output channel, in which acts as a feature map.

2.6 GCC-PHAT

GCC-PHAT is short for *Generalized Cross Correlation Phase Transform* and is a method for estimating the time difference of arrival between two signals captured by two microphones. As the name implies, GCC-PHAT uses the phase of the received signals to find the pure time difference. First, the weighted cross spectrum of the two signals is calculated as in equation 2.4. By then taking the inverse Fourier transform of the resulting cross spectrum, the GCC-PHAT correlation function is obtained in equation 2.5. [26]

Since GCC-PHAT normalizes the magnitude, only phase is used in the estimation, the method therefore focuses on estimating the pure ITD instead of correlating spectral cues.

The estimated time delay is then estimated through equation 2.6.

$$\hat{G}(H_L, H_R) = \frac{H_L H_R^*}{|H_L H_R^*|} \quad (2.4)$$

$$\hat{g}(H_L, H_R, \tau) = \mathcal{F}^{-1} \{ \hat{G}(H_L, H_R) \} (\tau) \quad (2.5)$$

$$\hat{\tau}(H_L, H_R) = \arg \max_{\tau} \hat{g}(H_L, H_R, \tau) \quad (2.6)$$

2.7 Hilbert Transform

The Hilbert transform refers to the relationship between the magnitude and phase of a spectrum in a minimum phase system. In this work, the Hilbert transform is used to reconstruct phase of the spectrum from interpolated magnitudes in section 3.4. By assuming that the poles and zeros of the spectrum Z-transform lie inside the unit circle, the system is minimum phase. Under this assumption, the logarithm of the magnitude and the phase of the spectrum are a Hilbert transform pair [27].

The minimum phase *cepstrum* h_{c_min} can then be computed according to equation 2.7, where N is the length of h_c [27].

The cepstrum refers to the inverse Fourier transform of the logarithm of the magnitude according to equation 2.8.

$$h_{c_min}[n] = \begin{cases} h_c[0], & n = 0 \\ 2h_c[n], & 0 < n < N/2 \\ h_c[N/2], & n = N/2 \\ 0, & \text{otherwise} \end{cases} \quad (2.7)$$

$$h_c[n] = \mathcal{F}^{-1} \{ \log |H(f)| \} \quad (2.8)$$

The Fourier transform of h_{c_min} then yields the reconstructed signal containing the minimum phase components according to equation 2.9.

$$H_{\min_phase} = \mathcal{F}\{h_{\text{cepstrum_min}}\} \quad (2.9)$$

2.8 Evaluation Methods

In this section the different evaluation methods are defined. The metrics aim to measure the accuracy of all localization cues.

2.8.1 Mean Square Error

Mean square error (MSE) measures the mean of the error squared and is defined according to equation 2.10. It will mainly be used in different loss functions in this paper.

$$\text{MSE}(\hat{H}, H) = \frac{1}{M} \sum_{k=1}^M (\hat{H}(k) - H(k))^2 \quad (2.10)$$

In this thesis, root mean square error (RMSE) is also used and is defined according to equation 2.11

$$\text{RMSE}(\hat{H}, H) = \sqrt{\frac{1}{M} \sum_{k=1}^M (\hat{H}(k) - H(k))^2} \quad (2.11)$$

2.8.2 Log Spectral Distance

Log spectral distance (LSD), is used to evaluate the accuracy of the used methods ILD and spectral cues.

LSD is defined according equation 2.12, where M is each frequency bin of the HRTF, $\hat{H}(k)$ is the estimated spectrum, and $H(k)$ is the true spectrum. LSD is the most commonly used metric for HRTF evaluation. [28]

$$\text{LSD}(\hat{H}, H) = \sqrt{\frac{1}{M} \sum_{k=1}^M \left(20 \log_{10} \frac{|\hat{H}(k)|}{|H(k)|} \right)^2} \quad (2.12)$$

2.8.3 Interaural Time Delay Error

Interaural time delay error (ITDE) measures the estimated time difference of arrival error of two ears using GCC-PHAT. This is the chosen metric to calculate the ITD error and it defined according to equation 2.13

$$\text{ITDE}(\hat{H}_L, \hat{H}_R, H_L, H_R) = \hat{\tau}(\hat{H}_L, \hat{H}_R) - \hat{\tau}(H_L, H_R), \quad (2.13)$$

where $\hat{\tau}$ refers to the ITD estimation in equation 2.6.

To measure the ITDE over several points, the root mean square ITDE (RMS-ITDE) is used and is defined according to equation 2.14

$$\text{RMS-ITDE}(\hat{H}_L, \hat{H}_R, H_L, H_R) = \sqrt{\frac{1}{N} \sum_{i=1}^N \text{ITDE}(\hat{H}_{L,i}, \hat{H}_{R,i}, H_{L,i}, H_{R,i})^2}, \quad (2.14)$$

where N is the number of points.

2.8.4 Interaural Level Difference Error

Interaural Level Difference error (ILDE), is used to measure the error of pressure difference between the ears. The ILDE error is defined according to equation 2.15:

$$\text{ILDE}(\hat{H}_L, \hat{H}_R, H_L, H_R) = \text{ILD}(\hat{H}_L, \hat{H}_R) - \text{ILD}(H_L, H_R), \quad (2.15)$$

where ILD refers to the ILD definition according to 2.2.

To measure the ILDE over several points, the root mean square ILDE (RMS-ILDE) is used and is defined according to equation 2.16

$$\text{RMS-ILDE}(\hat{H}_L, \hat{H}_R, H_L, H_R) = \sqrt{\frac{1}{N} \sum_{i=1}^N \text{ILDE}(\hat{H}_{L,i}, \hat{H}_{R,i}, H_{L,i}, H_{R,i})^2}, \quad (2.16)$$

where N is number of points.

2.9 Related Work

Earlier work has shown great results in the field of sonification.

One of the earlier works was the *SeeHear* system, developed by Nielsen et al. [3]. They built a *VLSI* (Very-large-scale integration) chip with a photodetector and a set of analog filters resulting in a system producing simplified localization cues for ITD, ILD and elevational cues for visual events. The resulting system allowed the user to perceive and localize the localization cues. However, compared to the advancements of technology through digitalization and the use of HRTFs, more accurate localization cues can be generated.

Due to the high degree of personalization of HRTF measurements and the cost and difficulty of the measurement setup, more recent work has explored personalized HRTF models that adapts the HRTF to an individual.

Andersen et al. [29] evaluated how individualized HRTFs can improve first person shooter games. They found that localization accuracy improved with HRTFs compared to non-binaural sound. The accuracy further improved with further personalized HRTFs.

Lee and Kim [30] proposed a model based on deep neural networks. The model was built on three subnetworks. Subnetwork A encodes ear, head, and torso anthropometrics. Subnetwork B is a CNN that encodes a Canny Edge Detected image of the ear. Subnetwork C is a network that fuses the two subnetworks A and B. The result yielded an increased localization accuracy compared to an average HRTF measurement.

Pirard [31] proposed another personalized model. Priards method uses a CNN trained on a landmark dataset to detect anatomical landmarks on different ear images. From these landmarks, 12 landmarks are selected to compute 7 anthropometric measurements. From the 7 anthropometric measurements, a nearest-neighbour search is done to find the best matching HRTF dataset.

3 Method

In this chapter, the different methods are defined and described. These methods are later used to define different solutions to the sonification problem and support the experimental setup.

3.1 Head-Related Transfer Function

The HRTFs are used to model the ITD, ILD, and elevation and antero-posterior cues as a function of direction.

To extract data from the .sofa files, the pysofaconventions *Application Programming Interface* (API) is used [32]. The API is used to collect certain data, this data and correlating API function calls are listed in table 3.1.

The HUTUBS database contains 96 subjects i.e HRTF datasets. Each subject is labeled with a number between 1 and 96, and has a corresponding .sofa file. The database also includes a file for anthropometric measures for subjects except subject 18, subject 79 and subject 92. In the database, two subjects have repeated measures, subject 1 and subject 96, and subject 22 and subject 88. Each subject dataset contains 440 points with correlating 256 sampled HRIRs measured from a distance of 1.47 m. The sample rate of these HRIRs are 44.1 kHz.

3.2 Distance filter

To create cues for distance perception, a distance filter H_{distance} is introduced. The filter is a combination between the intensity decrease $H_I(r, r_0)$ and the spectral attenuation $H_\alpha(f, \Delta r)$,

API call	Description
getDataIR()	Returns the impulse responses(HRIR) for all receivers.
getSamplingRate()	Returns the sampling rate of the receivers.
getVariableValue('SourcePosition')	Returns all emitter positions and is aligned with the impulse responses.
getVariableValue('ReceiverPosition')	Returns the receiver positions.

Table 3.1: Used calls in Pysofaconventions

where r is the target distance, r_0 is the sound reference distance, and $\Delta r = r - r_0$. H_{distance} is defined according to equation 3.1.

$$H_{\text{distance}}(f, r, r_0) = H_I(r, r_0) + H_\alpha(f, r - r_0), \quad [\text{dB}] \quad (3.1)$$

The intensity decrease filter $H_I(r, r_0)$ aims to model how the intensity changes over distance. A derivation of the filter is done in by using equation 2.3. By assuming the the reference sound has intensity I_0 and the desired sound has intensity I , the sound intensity difference ratio can be calculated through equation 3.2.

$$I_{\text{ratio}} = \frac{I}{I_0} = \frac{r_0^2}{r^2} \quad (3.2)$$

The resulting filter expressed in dB can the be calculated as in equation 3.3.

$$H_I(r, r_0) = 10 \cdot \log_{10}(I_{\text{ratio}}) = 20 \cdot \log_{10}\left(\frac{r_0}{r}\right), \quad [\text{dB}] \quad (3.3)$$

The spectral attenuation filter is a low-pass filter that models the atmospheric absorption. The filter is a simplification of the ISO standard ISO 9613-1:1993 *Acoustics - Attenuation of sound during propagation outdoors*[33]. It is a standard for how the sound attenuates as it travels through air outdoors. The simplification of the formula for the pure tone sound attenuation coefficients α [dB/m] of the standard can be seen in equation 3.4.

$$\begin{aligned} \alpha = & 8.686 f^2 (1.84 \times 10^{-11} + \\ & 0.01275 e^{\frac{-2239.1}{T_0}} (f_{rO} + \frac{f^2}{f_{rO}})^{-1} + \\ & 0.1068 e^{\frac{-3352}{T_0}} (f_{rN} + \frac{f^2}{f_{rN}})^{-1}), \quad [\text{dB/m}] \end{aligned} \quad (3.4)$$

In this simplified version, the ambient atmospheric pressure p_a is set to the same value as the reference ambient atmospheric pressure p_r . The ambient atmospheric temperature T is set to the same as the reference air temperature $T_0 = 293.15$ K. f_{r0} and f_{rN} are two relaxation frequencies defined according to equation 3.5 and 3.6.

$$f_{r0} = 24 + 4.04 \times 10^4 h \frac{0.02 + h}{0.391 + h} \quad (3.5)$$

$$f_{rN} = 9 + 280 h \quad (3.6)$$

In these two equations, h is the molar concentration of water vapour as a percentage. According to the standard, h ranges roughly between 0.2% and 2% near sea level altitudes. In this work, the concentration of water vapour was set constant to $h = 1\%$. This simplified version is now only dependent on one variable f , which is the desired frequency for which to calculate α .

With α , the absorption filter $H_\alpha(f, \Delta r)$ can be created according to equation 3.7 where Δr is the radius difference between the point of the reference sound and the point of the interpolated sound.

$$H_\alpha(f, \Delta r) = -\alpha(f) \Delta r \quad [\text{dB}] \quad (3.7)$$

3.3 Interpolation

As the HRTF datasets contain sampled points, some points need to be interpolated. In the project, three main solutions were tested. Two interpolation techniques and one where the closest point is used. The closest point function is only used for testing.

3.3.1 Bilinear interpolation

The first interpolation technique used is bilinear interpolation. This method uses a weighted averaging function of N points creating w_i weights for each point p_i . The averaging function used is an inverse distance function where the closest point contributes the most to the interpolated point. The interpolated impulse responses h_p are then calculated using the formula 3.8 h_i is the corresponding impulse response to each point p_i .

$$h_p = \sum_{i=1}^N \frac{w_i h_i}{\sum_{j=1}^N w_j} \quad (3.8)$$

As the points in the used HRTF dataset are on the sphere, this type of interpolation is only accurate for the points that are on that sphere with radius r . To solve this problem, each point $\mathbf{p}_{\text{proj}}$ that needs to be interpolated can be projected onto the sphere. By creating a directional vector $\mathbf{dV} = \mathbf{p} - \mathbf{p}_{\text{ear}}$ where $\mathbf{p}$ is the desired point to interpolate and $\mathbf{p}_{\text{ear}}$ is the position of the ear, a line L can then be defined according to equation 3.9.

$$L(t) = \mathbf{p}_{\text{ear}} + t \mathbf{dV}, \quad t \in \mathbb{R} \quad (3.9)$$

Then the distance to the point $\mathbf{p}_{\text{proj}}$ should be equal to r which implies that $L^T L = r^2$ in which gives the expression in equation 3.10.

$$r^2 = t^2 \mathbf{dV}^T \mathbf{dV} + 2t \mathbf{p}_{\text{ear}}^T \mathbf{dV} + \mathbf{p}_{\text{ear}}^T \mathbf{p}_{\text{ear}} \quad (3.10)$$

This is a normal scalar quadratic equation in t that will have two roots for $r > 0$, one positive root and one negative root. As the sought point $\mathbf{p}_{\text{proj}}$ is in the direction of $\mathbf{dV}$, the positive root is the desired value t_{proj} . Then the projected point $\mathbf{p}_{\text{proj}} = L(t_{\text{proj}})$.

3.3.2 Neural network

As each set of data in a HRTF dataset has many points on the sphere with correlating impulse responses, a neural network with supervised learning can be trained. The dataset used in the model is all the pairs of point and impulse response available in one dataset.

The neural network is a fully connected multilayer perceptron (MLP) that maps a spherical position to an impulse response. It has an input layer with two nodes, one representing the azimuth, and one representing the elevation. See figure 3.1. On its own, the input layer has low dimensionality and the encoder does not learn a continuous representation of the unit circle as $|\theta| = 180^\circ$ leads to a discontinuous wrap-around. To solve this problem, an embedding layer is added that transforms the angles by taking the sine and cosine of both angles. The result is a layer with the doubled amount of nodes and is continuous at the wrap-around angles. See figure 3.1 for visual representation for the embedding.

The radius is disregarded by the neural network due to all points are on the sphere and the radius is constant over all points. For a point not on that sphere, the point can not be interpolated by the neural network. Instead, when a point not on the sphere, it is projected on the sphere according to section 3.3.1. Then the distance filter from section 3.2 is used to include the distance cues in the HRTF.

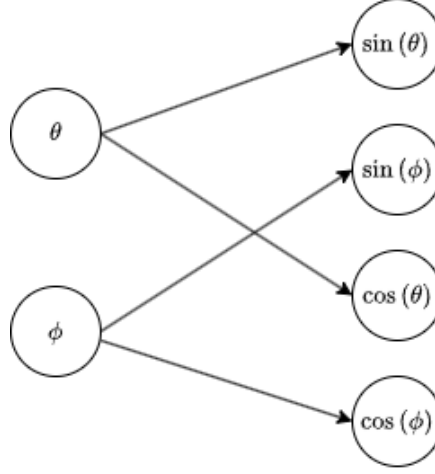


Figure 3.1: Positional embedding of positional input layer

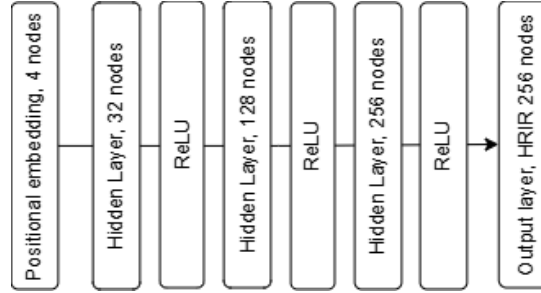


Figure 3.2: Architecture of HRIR neural network interpolation model

In this work two different neural networks are evaluated, one that utilizes the RMSE for HRIR as loss function and one that minimizes the RMSE for HRTF as loss function.

In the network for the HRIR loss function, the input layer is followed by 3 hidden layers of sizes [32, 128, 256] nodes. Each hidden layer is followed by a ReLU activation function. The last layer, the output layer, is the same size as the number of samples in the impulse responses in the used dataset. In this work, the size was 256 samples. See figure 3.2 for visualization.

In the network for the HRTF loss function, the network is split up into two subnetworks. The first network interpolates the single-sided magnitude of the HRTF while the other subnetwork interpolates the phase of the network. The magnitude consists of 5 hidden layers of sizes [16, 32, 64, 128, 256] and then the resulting output layer of the size of the real-sided magnitude. The subnetwork for the phase has 3 hidden layers of sizes [32, 128, 256] and the resulting output layers. The raw wrapped phase has a sawtooth-like appearance, which is hard for a neural network to predict. Instead of using raw phase, the phase is unwrapped. Unwrapping the phase yields almost linear function instead of the sawtooth-like pattern, which is much easier for the network to predict. See figure 3.3 for visualization of the HRTF neural network.

Both models were trained with a learning rate of 10^{-3} for 10^4 epochs and with a batch size of 16.

3.4 Personalized HRTF

As the HRTFs are dependent on the shape of the ear and head, the directional HRTFs are heavily individual. A personalized model to predict the personalized HRTF is developed and trained. The personalized model is a continuation of the model in section 3.3.2 and in-

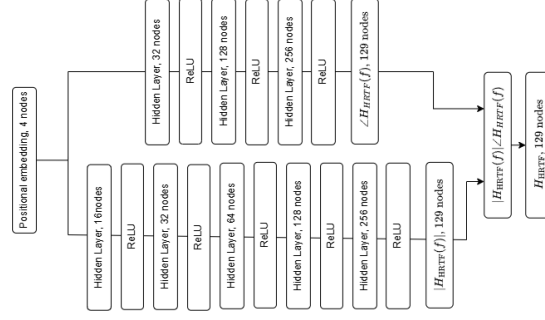


Figure 3.3: Architecture of HRTF neural network interpolation model

spired by Lee and Kim’s network [30]. Compared to the model in section 3.3.2, personalized anthropometric ear and head measurements are introduced. There are many ways to measure the anthropometrics. Therefore, a standard for anthropometric data collection is needed for meaningful predictions. In this work, the HUTUBS definition is used [19][20]. In this definition, 15 of the metrics were used and are listed in table 3.2 and are visualized in figure 3.5 and 3.4.

Metric	Description	Used
x_1	head width	x
x_2	head height	x
x_3	head depth	x
x_4	pinna offset down	
x_5	pinna offset back	
x_6	neck width	
x_7	neck height	
x_8	neck depth	
x_9	torso top width	
x_{12}	shoulder width	
x_{14}	height	
x_{16}	head circumference	
x_{17}	shoulder circumference	
d_1	cavum concha height	x
d_2	cymba concha height	x
d_3	cavum concha width	x
d_4	fossa height	x
d_5	pinna height	x
d_6	pinna width	x
d_7	intertragal incisure	x
d_8	cavum concha depth (down)	x
d_9	cavum concha depth (back)	x
d_{10}	crus of helix	x
θ_1	pinna rotation angle	x
θ_2	pinna flare angle	x

Table 3.2: Used anthropometrics from HUTUBS[19][20]

As the localization cues are dependent on different anthropometrics, the network was split into different parts. The ITD and ILD are mainly dependent on the head sizes x_1 - x_3 while the elevation and antero-posterior cues are mainly dependent on the pinna shape d_1 - d_{10} and pinna angles θ_1 and θ_2 . This creates a three part network where each subnetwork is responsible for a specific localization cue and with a specific input. However, implementing the three part network is not straightforward. The ILD, elevation, and antero-posterior cues

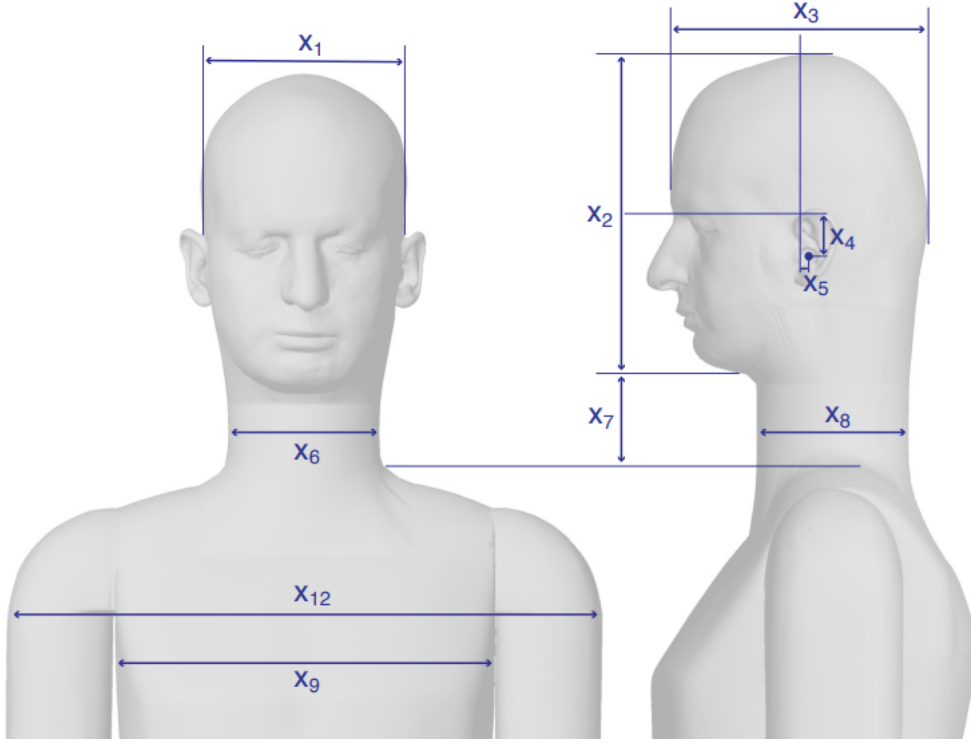


Figure 3.4: Visualization of head and torso anthropometrics (Source: HUTUBS[19][20], CC BY 4.0)

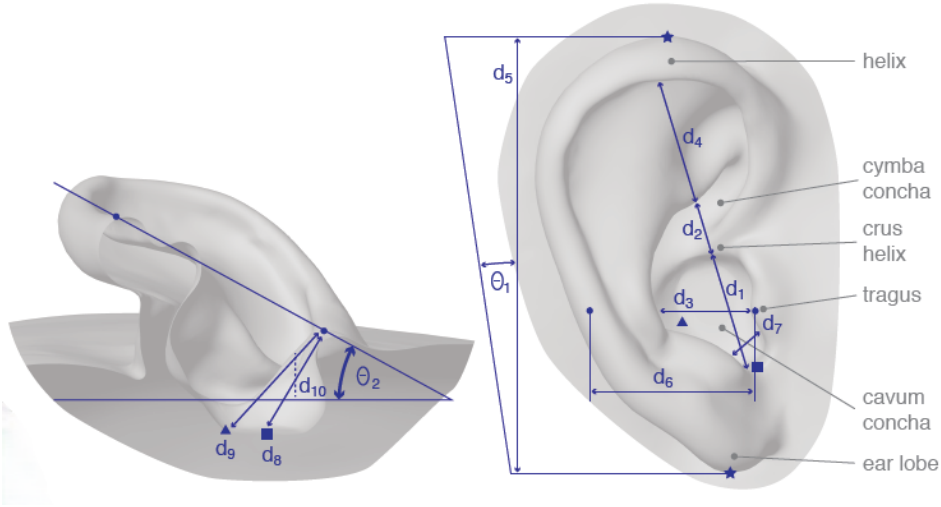


Figure 3.5: Visualization of ear anthropometrics (Source: HUTUBS[19][20], CC BY 4.0)

are all combined in the magnitude of the signal. Therefore, it is not straightforward to split the spectral attributes of these cues. Instead, the network is split into two subnetworks instead. The first subnetwork predicts the ITD τ_{ITD} and the second subnetwork predicts the magnitude $|H_{HRTF}(f)|$. The resulting network architecture is visualized in figure 3.6.

The resulting network will not be able to model the phase properties created by pinna reflections and other reverberations as we only focus on the pure ITD.

Reconstructing the HRTF in this manner with linear phase has both advantages and drawbacks. The biggest drawback is that timing of arrival for different frequencies. The timing differences occur due to reflections in the pinna which is part of the spectral cues used by the

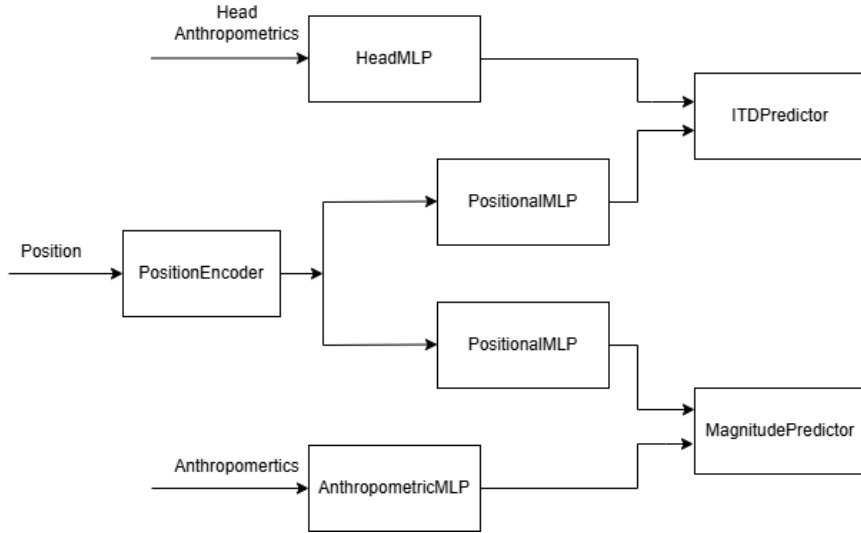


Figure 3.6: Architecture of the personalized network

brain for localization. However, the reflection in the pinna acts like a filter, creating frequency notches and peaks in the magnitude. So the pinna reflections are also modelled in the magnitude. The advantages, on the other hand, are that information from all reflections is excluded from the phase. The reflections are hard to model, and removing the phase decreases training times. As a result, the focus is on reconstructing the HRTF phase with an accurate ITD prediction and ease localization rather than physical accuracy of the HRTF.

3.4.1 Positional MLP

The Positional MLP encodes the positional embedding created by the positional encoder in figure 3.1 from section 3.3.2. The four node embedding is the input layer of the MLP and is followed by 3 hidden layers of [16, 32, 32] nodes and then by an output layer of 16 nodes. The Positional MLP is visualized in figure 3.7

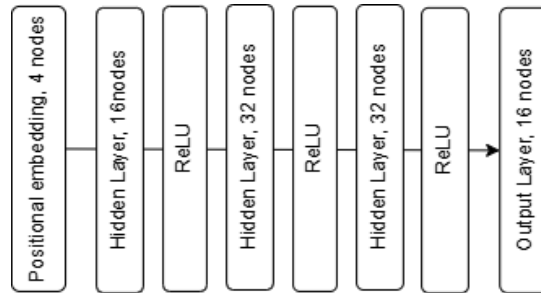


Figure 3.7: Architecture of the Positional MLP

3.4.2 Anthropometric MLP

The anthropometric MLP is a normal MLP that takes the anthropometrics as input layer. The input layer is then followed by 2 hidden layers of sizes [16, 32] and then an output layer of 16 nodes. The architecture is visualized in figure 3.8

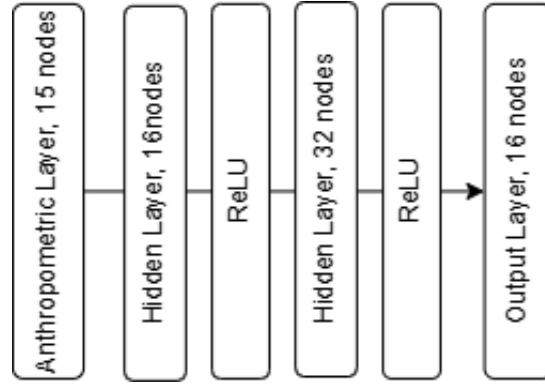


Figure 3.8: Architecture of the Anthropometric MLP

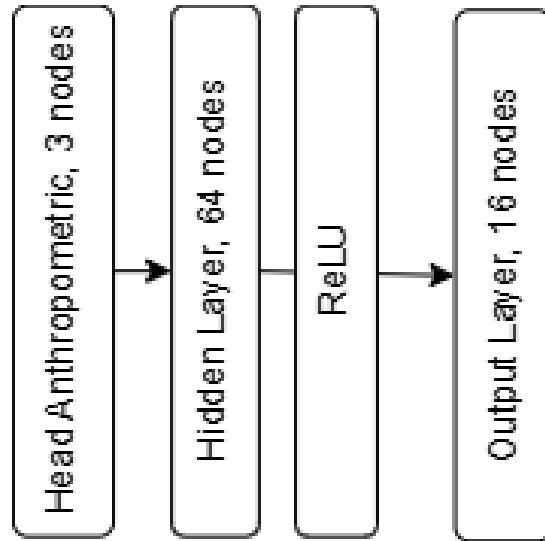


Figure 3.9: Architecture of the Head MLP

3.4.3 Head MLP

The Head MLP takes the head metrics x_1 - x_3 as input layer and is then followed by a hidden layer of 64 nodes and an output layer of 16 nodes. The architecture is visualized in figure 3.9

3.4.4 ITD Predictor

The ITD predictor is the fusion between a Positional MLP and the Head MLP. The input layer is the Positional MLP output layer and Head MLP layer stacked on top of each other creating a layer of 64 nodes. The input layer is followed two hidden layers with [64, 32, 16] nodes. The output layer consists of one node representing the ITD. The architecture is visualized in figure 3.10

3.4.5 Magnitude Predictor

The Magnitude predictor is a fusion between a Positional MLP and the Anthropometric MLP. The input layer is the Positional MLP and Anthropometric MLP output layers stacked on top of each other creating an input layer of 32 nodes. The input layer is then followed by 3 hidden layers of sizes [64, 128, 129]. The three layers are then followed by a CNN with two layers of kernel size of 7 and a padding of size 3. The first layer expands the input channel to 4

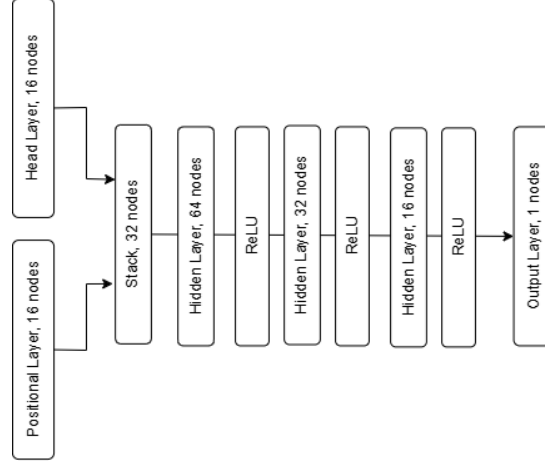


Figure 3.10: Architecture of the ITD Predictor

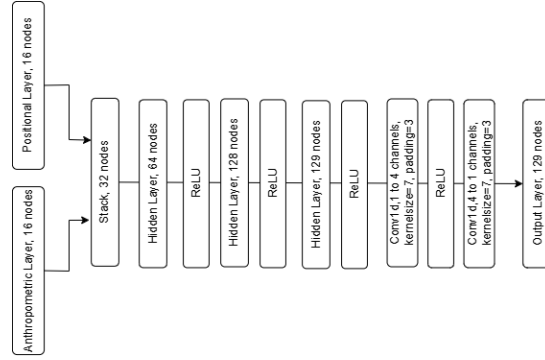


Figure 3.11: Architecture of the Magnitude Predictor

channels and the other brings it back to 1 channel and the resulting magnitude $|H_{\text{HRTF}}(f)|$. The architecture is visualized in figure 3.11

3.4.6 HRTF Reconstruction

To reconstruct the HRTF, the Hilbert transform is used to calculate the minimum phase spectrum from the predicted magnitude according to section 2.7 resulting in the signal $H_{\text{HRTF_min-phase}}$. This operation ensures the HRTF for both ears are causal. The farthest ear is then delayed with the predicted ITD according to equation 3.11.

$$H_{\text{HRTF}}(f) = H_{\text{HRTF_min-phase}}(f)H_{\text{ITD}}(f) \quad (3.11)$$

As τ_{ITD} is a scalar, it can be interpreted as a Dirac pulse in the time domain. This results in the pulse $h_{\text{ITD}}(t) = \delta(t - \tau_{\text{ITD}})$. Using the Fourier transform, the resulting frequency domain time delay H_{ITD} is calculated in equation 3.12.

$$H_{\text{ITD}}(f) = \mathcal{F}\{h_{\text{ITD}}(t)\} = \mathcal{F}\{\delta(t - \tau_{\text{ITD}})\} = e^{-2j\pi f\tau_{\text{ITD}}} \quad (3.12)$$

As the magnitude of $H_{\text{ITD}} = 1$ for all f , all information of H_{ITD} the signal is in the phase.

3.4.7 Supervised learning

When training model, supervised learning is used. The data used are from the HUTUBS dataset [19][20]. All subjects are included from the dataset except subject 1, subject 18, sub-

ject 22 subject 79, and subject 92. Subject 1 and subject 96 are double measurement, which means that the measurements are made on the same physical subject. Therefore, subject 1 is removed to avoid any potential model bias. The same is done for subject 22 as it is a double measurement together with subject 88. Subject 18, subject 79, and subject 92 are removed as they lack anthropometric data.

After the subject exclusion, all included subjects in the dataset are split into training and test sets depending on what experiment is done, see section 3.5.

For the Magnitude predictor, each subject contains measurements for two ears, the left and right ear. Instead of training on left ears and right ears separately, the right ear points are horizontally mirrored by using $-\theta$ instead of θ . This allows the right ear HRTFs to be treated as left ear HRTFs and the training data for each ear of the head is effectively doubled. While the ears of a subject are heavily correlated, they are not equal, i.e. they have different anthropometrics. As the ears of a subject are correlated, it is important to split the subject ears after the split into training, test and validation data. The optimal loss function for the magnitude predictor is to minimize the LSD for each batch. However, using pure logarithmic loss function is not a good idea. As magnitude decreases for higher frequencies due to the headshadow effect, the linear error becomes small for higher frequencies. However, in the logarithmic domain, the loss explodes for small linear errors. The exploding loss creates two main problems. The first problem is that the model only focuses on learning the higher frequencies instead of the overall shape of the magnitude and learning ILD cues. The second more severe problem is training becomes unstable at initialization due to the exploding loss. The result is that the optimizer stagnates, the model does not learn, and the loss remains constant. To solve the problems, two measures were made.

The first measure is that a linear MSE loss function L_{MSE} , is used in combination with a logarithmic MSE loss function L_{log} instead of LSD. Instead of using a constant loss function over all epochs, the first epoch only uses L_{MSE} to learn the overall shape of the magnitude before learning high frequency magnitude detail. For each epoch after the first epoch, L_{log} is slowly introduced up to epoch N_t by using a linearly increasing scaling factor α . In this work, epoch N_t was set to 100, and the scaling factor α starts at 0 and increases to 0.5

The second measure is assuming all magnitudes lower than ϵ is equal to 0. This is done through the pytorch method *clamp*. Clamp modifies the magnitude by replacing all magnitudes lower than ϵ with ϵ . In this work, ϵ was set to 10^{-8} but it can be increased as long as all notches in the magnitude are included.

The result is a more stable training that does not risk stagnating and ensures convergence. The loss function can be seen in algorithm 3.13

$$\begin{aligned} L_{\text{log}} &= \text{MSE}(\log_{10}(\text{clamp}(\hat{y}, \min = \epsilon)), \log_{10}(\text{clamp}(y, \min = \epsilon))) \\ L_{\text{MSE}} &= \text{MSE}(\hat{y}, y) \\ L_{\text{MAG}}(e) &= \begin{cases} L_{\text{MSE}} + \alpha_{N_t} L_{\text{log}}, & e > N_t \\ L_{\text{MSE}} + \alpha_e L_{\text{log}}, & e < N_t \end{cases} \end{aligned} \quad (3.13)$$

For the ITD predictor, for each subject and position, the time delay between the two ears is calculated through GCC-PHAT. For each batch, the loss is calculated through equation 3.14, where N_b is the batch size, $\hat{\tau}$ is the predicted ITD and τ is the true ITD

$$L_{\text{ITD}} = \frac{1}{N_b} \sum_{k=1}^{N_b} (\hat{\tau}_k - \tau_k)^2 \quad (3.14)$$

During training, both models were trained with a batch size of $N_b = 16$ together with a learning rate of 5×10^{-4} . For each batch, small white noise is added to the anthropometrics with standard deviation of 0.1 mm. The models were trained for 300 epochs each.

3.5 Experimental Setup

In this section, the experimental setup is defined and labeled. It describes the idea behind the experiment, what methods will be included and what the experiment is meant to measure.

3.5.1 Interpolation Experiment

In the Interpolation experiment, the different HRTF interpolation methods will be evaluated analytically. The used methods that will be compared are:

1. Nearest Neighbor (Nearest)
2. Bilinear Interpolation (Bilinear)
3. HRIR neural network (nn_{HRIR})
4. HRTF neural network (nn_{HRTF})

In this experiment, the points in the dataset corresponding to subject 1 are split into two different subsets, a training set, and a test set, in a 90–10% split. For each point in the test set, each approach interpolates the validation point using the training set. The neural networks were trained for 10000 epochs on the training set. To measure the results, LSD and ILD error is used to calculate the spectral error, and GCC-PHAT is used to find the pure time delay error. Six points are randomly chosen from the validation set. All interpolated magnitudes for those points are plotted together with the true magnitude.

3.5.2 Personalized HRTF Prediction vs Dataset Selection

In this experiment, the model for personalized HRTF is compared against HRTF dataset selection. Subject 2, subject 3, and subject 4 in the HUTUBS dataset are used as test data. The rest of the data within the HUTUBS database are used as training data. For each subject in the training data, each dataset is compared to each test dataset. This is done by taking iterating over all positions and calculating the LSD, RMS-ILDE and RMS-ITDE for that position. For each training subject, the mean LSD, RMS-ILDE, and RMS-ITDE is calculated. Moreover, the lowest LSD, RMS-ILDE, and RMS-ITDE is also calculated. For each subject and position in the test dataset, the personalized model interpolates a HRTF and the LSD, RMS-ILDE, and RMS-ITDE is calculated and averaged for all test subjects. Six points are randomly chosen. For each subject in the test set, the six chosen points corresponding magnitudes are plotted together with the predicted magnitude and the best matching subjects magnitude.

The LSD, RMS-ILDE, and RMS-ITDE calculation is done on 9 other models. For each of these models, the number of training subjects used for training in the personalized network are varied. The first model use 10% of the training data. For each new model, 10 percentile points more of the data is added. This is done by shuffling the subjects using the python module *random* and setting *random.seed(32)*. For the first model, the subjects correlated to the first 10% of the list is used for training. For each model after the first model, more and more data is added. The resulting LSD, RMS-ILDE, and RMS-ITDE for each model is then plotted over the proportion of used training data. This is then compared with the average-matching selected subject and best matching selected subject.

3.5.3 Anthropometric Feature Reduction

In this experiment, the goal is to minimize the input anthropometric layer x in the Anthropometric MLP by only including the most essential metrics. The motivation behind it is to find a model that ease measurement extraction as much as possible while maintaining good localization accuracy. The HUTUBS subjects are divided into 90-10% training and test data using the python module *random* and setting *random.seed(32)*.

	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$
x_1	x	x	x	x	x
x_2	x	x	x	x	x
x_3	x	x	x	x	x
d_1	x	x	x	x	x
d_2	x	x		x	
d_3	x	x	x	x	x
d_4	x	x	x	x	x
d_5	x	x	x	x	x
d_6	x	x		x	
d_7	x	x		x	
d_8	x		x	x	x
d_9	x		x	x	x
d_{10}	x		x	x	x
θ_1	x	x	x		
θ_2	x	x	x		

Table 3.3: Definition of all input layers for the Anthropometric MLP in the anthropometric feature reduction experiment.

Five different models with different input layers are evaluated. For each subject in the test dataset, each model predicts the magnitude. The resulting LSD is calculated and then the mean LSD over all subjects for each model are presented. Furthermore, the average ILD error over the test subjects is also calculated.

The first model input layer $\mathbf{x}_1$ uses all anthropometrics from 3.2. This layer acts as a measurement baseline for the other models.

In the second model input layer $\mathbf{x}_2$, the metrics d_8 - d_{10} are removed. The motivation behind removing these metrics is that these metrics can not be measured on the pinna, instead these needs to measured from inside the cavum concha. Removing these metrics would greatly ease measurements.

In the third model input layer $\mathbf{x}_3$, the metrics d_2 , d_6 , and d_7 are removed. These metrics are chosen due to the hypothesis that these three metrics are the least significant.

In the fourth model input layer $\mathbf{x}_4$, the metrics θ_1 , and θ_2 are removed. Removing these two metrics makes sure all anthropometrics are in the same unit (cm), as θ_1 and θ_2 are angles expressed in degrees.

In the fifth model input layer $\mathbf{x}_5$ is a combination of the third and fourth model, removing d_2 , d_6 , d_7 , θ_1 , and θ_2 are removed. The different anthropometric input layers are listed in table 3.3.

4 Results

This chapter is dedicated to present the results. For each experiment, the result is presented through tables and figures. These are then interpreted and correlated with each other.

4.1 Interpolation Experiment

In table 4.1, the resulting LSD, RMS-ITDE, and RMS-ILDE is presented for each used method in the interpolation experiment.

	Nearest	Bilinear	nn _{HRIR}	nn _{HRTF}
LSD [dB] (Eq 2.12)	4.14	9.99	7.67	5.59
RMS-ITDE [μ s] (Eq 2.14)	64.7	43.8	85.4	42.1
RMS-ILDE [dB] (Eq 2.16)	2.43	2.14	1.34	0.65

Table 4.1: Results of HRIR interpolation from the interpolation experiment

Looking at the LSD, the bilinear interpolation has the highest LSD of 9.99 dB. In theory, this is expected. Bilinear interpolation averages the spectrum between 4 points. Averaging between points suppresses higher frequencies and acts as a lowpass filter. As the higher frequencies are suppressed, spectral cues located in higher frequency spans get attenuated and blurred as each point correlates to individual sharp peaks and notches. Although the LSD is high, the RMS-ILDE relatively low at 2.14 dB compared to the nearest neighbour RME-ILDE of 2.43 dB. RMS-ITDE for the bilinear interpolation is 43.8 μ s, in which is very low. As the RMS-ITDE and RMS-ILDEs are low, it gives the clue that bilinear interpolation works well for horizontal localization. From a more practical standpoint, the bilinear interpolation actually plays the sound from 4 different positions at different volumes instead of accurately playing the sound from one point.

The HRIR neural network has the second highest LSD of 7.67 dB and the worst RMS-ITDE of 85.4 μ s. This result is very expected as the neural network aims to minimize the MSE loss for the time-domain HRIR. The resulting predicted impulse response errors in time domain might be small, but small deviations in time-domain may have large impacts on the spectral features of the signal. Although the spectral inaccuracy, the RMS-ILDE of 1.34 dB is very low.

The HRTF neural network has the second lowest LSD of 5.59 dB, the lowest RMS-ITDE of 42.1 μ s and the lowest RMS-ILDE of 0.65 dB. The strong improvement compared to the HRIR

neural network is due to the loss function now aims to minimize the important features of the spectrum, the magnitude and the phase.

Although the HRTF neural network is an improvement, the closest point method has the lowest LSD of 4.14 dB. However, the RMS-ITDE for the nearest neighbour of $64.7 \mu\text{s}$ is very high, which is expected as no effort is done to improve the timing of the filters. The RMS-ILDE is also high with 2.43 dB.

In figure 4.1, the magnitude of 6 interpolated positions are plotted against the true magnitude. The blue lines represent the true magnitude, the orange lines represent the closest point, the green line represent the bilinear interpolation, the red line represent the HRIR neural network, and the purple line represent the HRTF neural network. The example positions plotted in figure 4.1 are:

- $(\theta_1, \phi_1) = (-20^\circ, 60^\circ), (\theta_2, \phi_2) = (132^\circ, -40^\circ)$
- $(\theta_3, \phi_3) = (-96^\circ, -40^\circ), (\theta_4, \phi_4) = (-140^\circ, -10^\circ)$
- $(\theta_5, \phi_5) = (-70^\circ, -10^\circ), (\theta_6, \phi_6) = (-0^\circ, 90^\circ)$

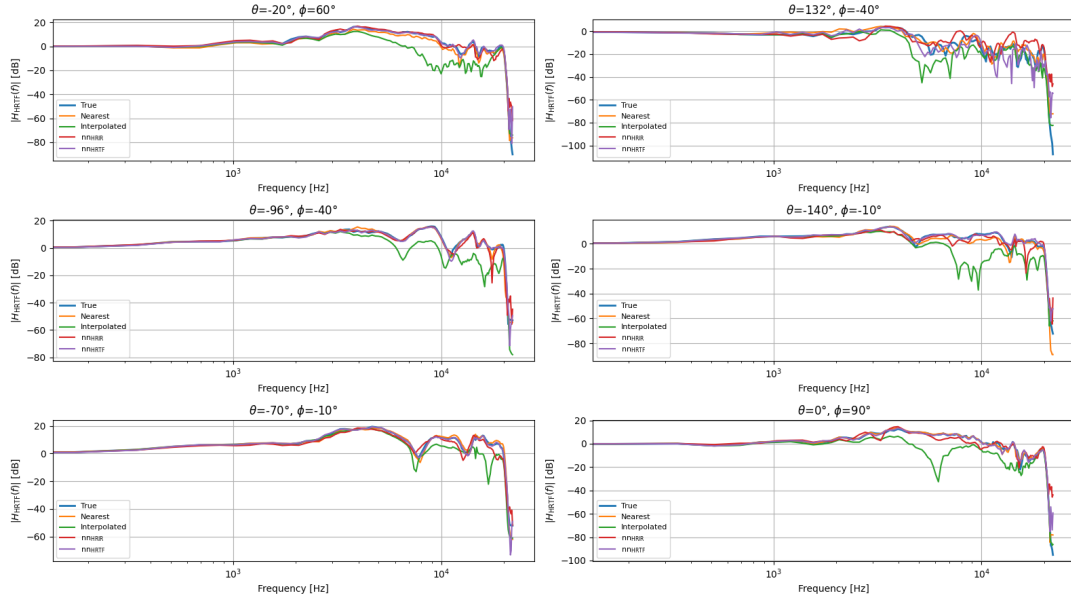


Figure 4.1: Result of interpolated magnitudes plotted against the true magnitude for six directions.

The plots in figure 4.1 confirms the results from table 4.1, but also gives a better picture of the method performance. Starting with the bilinear network, the trend is very clear. At lower frequencies, the interpolated magnitude follows the true magnitude well. However, as the frequency increases, the magnitude deviates more from the true magnitude which confirms the theory from table 4.1. However, comparing the notches and peaks between the true magnitude and interpolated magnitude, there are many notches where there are not suppose to be notches. Furthermore, there are notches in the wrong frequency bin and there are notches that the interpolation does not capture. The peaks are somewhat interpolated, but not necessarily accurate. The consequence of the bad high frequency notches and peaks is that the elevational, and antero-posterior cues are inaccurate and destroyed.

Continuing with the HRIR neural network and the red line in the plots. Although the LSD is relatively high, the magnitude follows the overall structure of the true magnitude. Although the magnitude is not perfect, most of the notches and peaks are correctly placed on

the frequency axis giving correct elevational, and antero-posterior cues. However, the model seems to struggle for the highest frequencies. For example, there is a big, wrongly predicted notch in the subplot (θ_3, ϕ_3) as well as in subplot (θ_4, ϕ_4) .

The HRTF neural network follows the true spectrum very closely for all frequencies. There are some outliers though, subplot (θ_2, ϕ_2) seems very hard to predict for all models, especially for the HRTF neural network.

In theory, the closest point method should follow the true magnitude well but have some misplaced, stronger, and/or attenuated notches and peaks on the frequency axis. The theory is true and visible in the plots. A good example is subplot (θ_3, ϕ_3) where there is a notch around 17 kHz that is wrongly placed, as well as the last peak that is shifted left on the frequency axis.

4.2 Personalized HRTF Prediction vs Dataset Selection

The main objective of this experiment is to measure how the personalized model perform. In table 4.2, the measurements for each subject is listed. For each subject, the mean performing dataset, the best performing dataset and the performance of the predicted HRTF using the personalized model is listed. For all points, the mean LSD, RMS-ITDE, and ITD error is listed.

Subject	Subject 2	Subject 3	Subject 4	Average
Mean error over all datasets				
LSD [dB]	7.67	7.81	7.69	7.72
RMS-ITDE [μ s]	49.5	49.1	49.6	49.4
RMS-ILDE [dB]	1.63	2.40	1.86	1.96
Best fitted HRTF				
LSD [dB]	6.21	6.18	6.16	6.18
RMS-ITDE [μ s]	36.4	35.7	35.8	36.0
RMS-ILDE [dB]	0.93	1.15	1.01	1.03
Predicted with Network				
LSD [dB]	6.12	5.93	5.93	5.99
RMS-ITDE [μ s]	118	43.6	45.0	68.5
RMS-ILDE [dB]	1.66	2.22	2.20	2.03

Table 4.2: Results of comparing HRTFs with Personalized model (LSD Eq 2.12, RMS-ITDE Eq 2.14, RMS-ILDE Eq 2.16).

Starting with the magnitude errors in the table, the predicted magnitudes LSD is very low for all subjects. For subject 2 the average LSD is 6.12 dB. For subject 3 and 4, the average LSDs are 5.93 dB. For all subjects, the personalized neural network outperforms the best fitted HRTF dataset where the LSD errors were 6.21, 6.18, and 6.16 dB.

The RMS-ITDE is very high compared to the best performing HRTF. Subject 2 has a RMS-ITDE of 118 μ s which is very bad. For subject 3 and 4, the RMS-ITDEs are 43.6 μ s and 45.0 μ s, these are better and more reasonable. Meanwhile, the best performing dataset has a RMS-ITDE of 36.4, 35.7, and 35.8 μ s.

The RMS-ILDEs are relatively high compared to the fitted datasets. The personalized model fails to score a lower RMS-ILDE than the mean performing fitted dataset for subject 2 and subject 4. For subject 2 the RMS-ILDE of 1.66 dB for the personalized model is close the the mean of 1.63 dB. However, for subject 4, the difference is larger. The personalized model scored an RMS-ILDE of 2.20 dB meanwhile the mean performing dataset has an RMS-ILDE of 1.86 dB. For subject 3, the personalized model got a lower RMS-ILDE of 2.22 dB compared to the mean performing dataset of 2.40 dB. However, the RMS-ILDE is far from the best fitted dataset where the RMS-ILDE is closer to 1 dB.

In figure 4.2, 4.3, and 4.4 the true magnitude for the left ear is plotted against the best fitted magnitude and the predicted magnitude for subject 2, 3, and 4 respectively. The blue line represent the true magnitude, the orange line represent the predicted magnitude, and the green line represent the best fitted HRTF magnitude. In each figure, there are six subplots, each subplot correlates to a direction. The positions are:

- $(\theta_1, \phi_1) = (130^\circ, -10^\circ)$, $(\theta_2, \phi_2) = (70^\circ, -10^\circ)$
- $(\theta_3, \phi_3) = (-30^\circ, 30^\circ)$, $(\theta_4, \phi_4) = (60^\circ, -30^\circ)$
- $(\theta_5, \phi_5) = (-10^\circ, 20^\circ)$, $(\theta_6, \phi_6) = (-60^\circ, -10^\circ)$

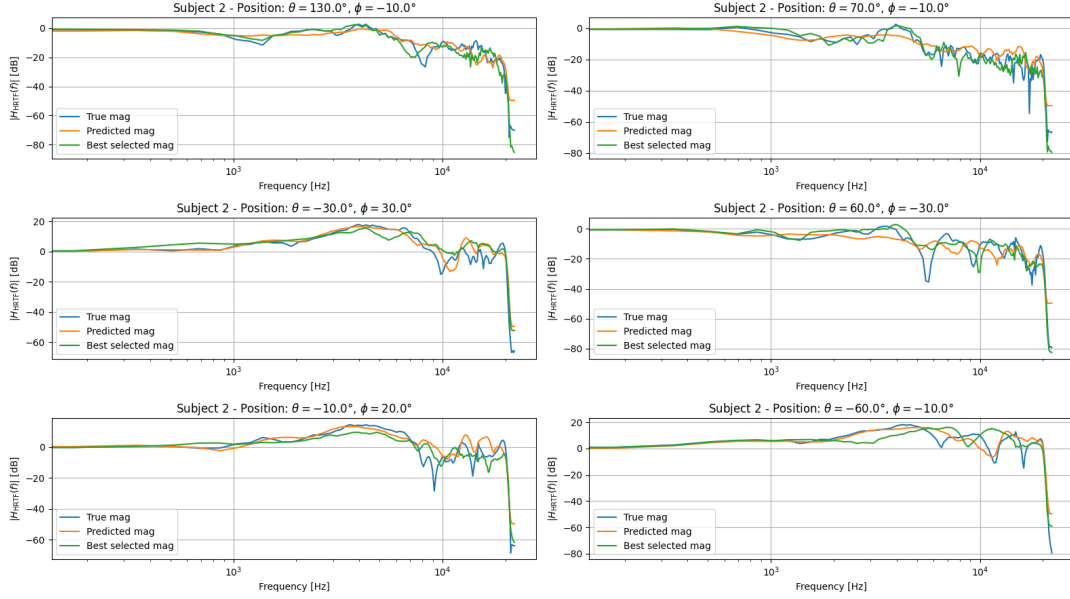


Figure 4.2: The best and predicted magnitude for 6 positions for subject 2

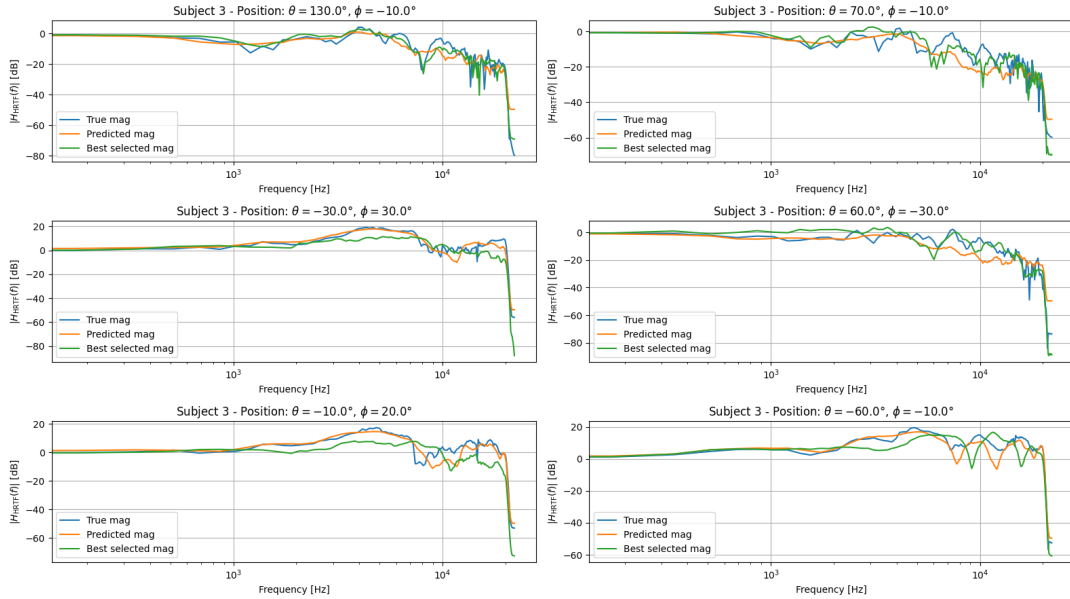


Figure 4.3: The best and predicted magnitude for 6 positions for subject 3

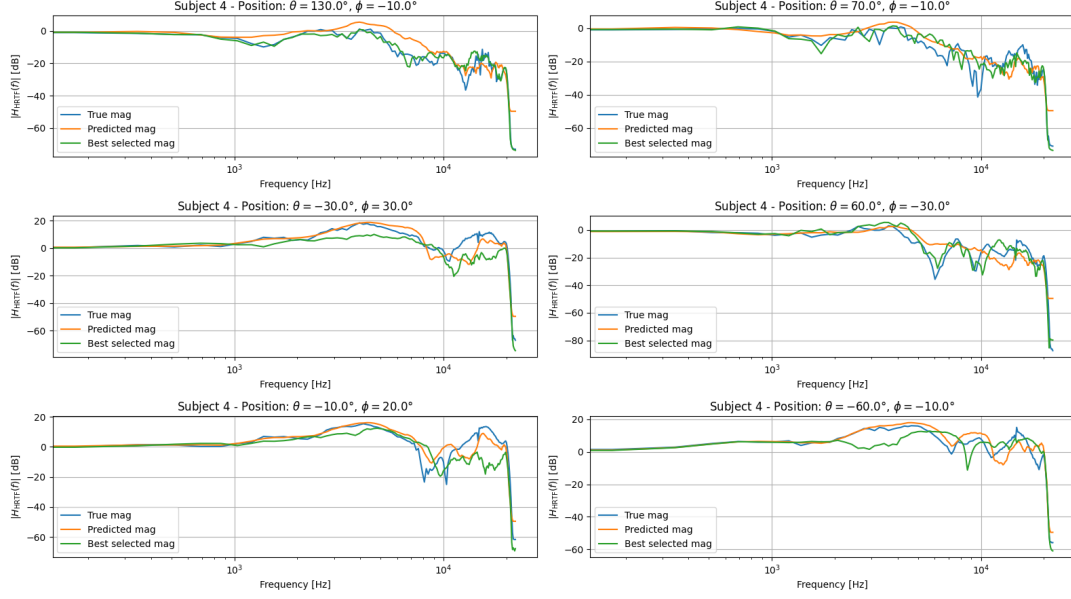


Figure 4.4: The best and predicted magnitude for 6 positions for subject 4

Looking at the magnitude plots for subject 2 in 4.2, the overall structure of the magnitude generally follows the structure of the true magnitude. However, the peaks and notches seems to be suppressed more the best performing HTRF. On the other hand, some of the best performing HRTF dataset notches seems to be more misplaced on the frequency axis in comparison to the predicted magnitude. In subject 2, both methods have strengths and weaknesses and perform similarly.

Comparing the magnitudes for subject 3 in figure 4.3, the trend is very similar to the result in the magnitude plots for subject 2. Many notches and peaks are suppressed by the personalized model. However, the overall structure is good. Meanwhile, the best fitted HRTFs notches are misplaced.

The magnitude comparison for subject 4 in figure 4.4 has the same trend as for subject 2 and 3. Even though the personalized model learned the overall structure very well and scores very low LSD, the notches and peaks, especially notches, gets suppressed by the model leading to suppressed localization cues.

In figure 4.5 and 4.6, the amount of training data for the personalized network is varied. For each trained model, the LSD and RMS-ITDE is plotted against the best fitted dataset and the mean performing fitted dataset. Each subplot represents a different subject and the bottom right subplot is the mean over all three subjects. The blue line represents the personalized model, the green line represents the mean performing dataset, and the orange line represents the best performing model.

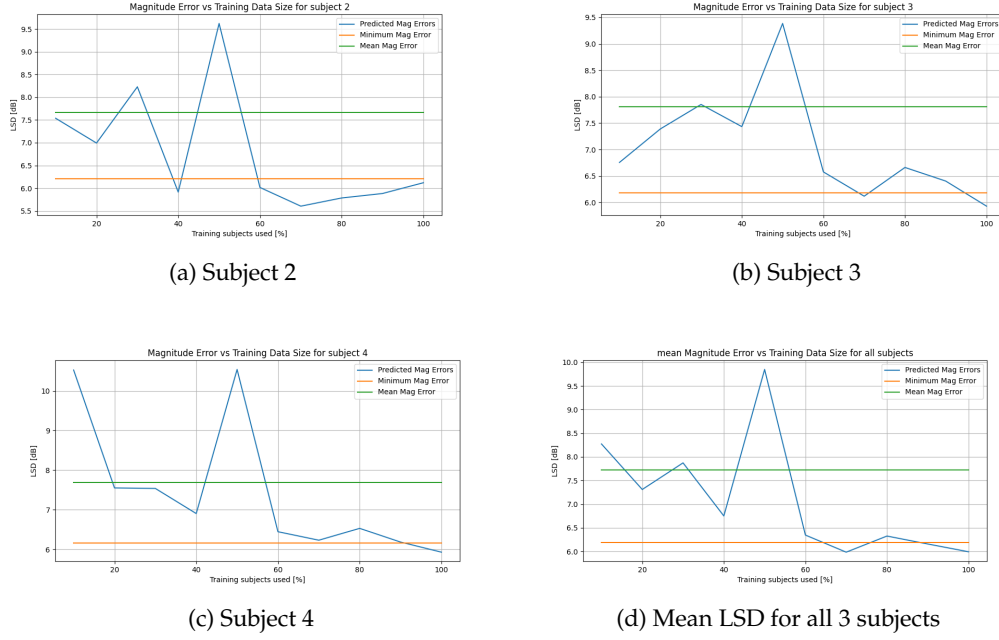


Figure 4.5: LSD for different amount of training data against best and mean magnitude

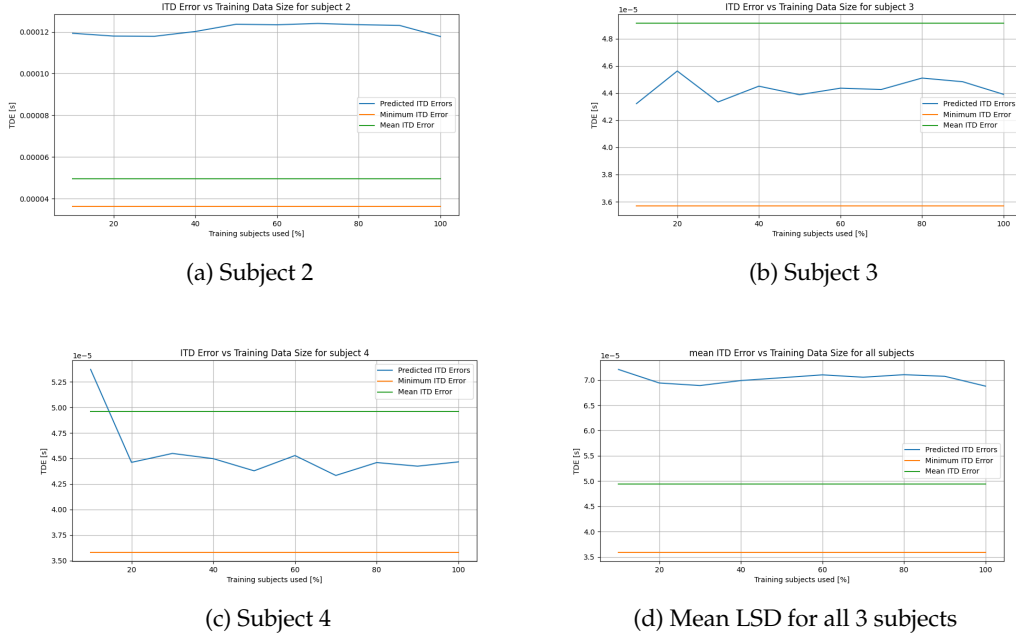


Figure 4.6: TDE for different amount of training data against best and mean magnitude

Looking at figure 4.5, the trend is very visible, increasing the training data, improves the models ability to predict the magnitude. However, for the case where the model was trained on 50% of the available data, there is a clear peak in the LSD. Comparing the predictions to the best fitted dataset and the mean performing dataset. The model quickly outperforms the mean performing dataset and then converges towards the best fitted dataset.

Looking at figure 4.6, the RMS-ITDE does not seem to improve much when introducing more data to the model. For subject 2, it is constantly high and does not improve when

more training data is introduced. For subject 3 and 4 however, the ITD is between the mean performing dataset and best fitted dataset.

4.3 Anthropometric Feature Reduction

The LSDs for each anthropometric input layer is listed in table 4.3. The best performing model is the model with input layer x_4 with an average LSD of 6.07 dB, the worst performing model is the model with input layer x_5 with an average LSD of 6.71 dB.

Used input layer	x_1	x_2	x_3	x_4	x_5
LSD [dB] (Eq 2.12)	6.45	6.22	6.29	6.07	6.71
RMSILDE [dB] (Eq 2.16)	2.06	2.15	1.94	1.98	1.99

Table 4.3: Results of different anthropometric input layers in the anthropometric feature reduction experiment

The lowest LSD of 6.07 dB was achieved by the model with input layer x_4 . The second lowest LSD of 6.22 dB was achieved by the model with input layer x_2 . The third lowest LSD of 6.29 dB was achieved by the model with input layer x_3 . The second highest LSD of 6.45 dB was achieved by the model with input layer x_1 . The highest LSD of 6.71 dB was achieved by the model with input layer x_5 .

For the RMS-ILDEs, all models perform very similar. The lowest RMS-ILDE was achieved by the model with input layer x_3 with in RMS-ILDE of 1.94 dB. Meanwhile the highest RMS-ILDE was achieved by the model with input layer x_2 with an RMS-ILDE of 2.15 dB

In figure 4.7 and figure 4.8, all models predicted magnitude for both ears are plotted against the true magnitude for two directions. Figure 4.7 plots the magnitudes for the direction $(\theta_1, \phi_1) = (72^\circ, 70^\circ)$ and figure 4.8 plots the magnitudes for the direction $(\theta_2, \phi_2) = (-100^\circ, 30^\circ)$.

The magnitude plots in figure 4.7 all follows the true magnitude closely to about 10 kHz. The differences are between 10 kHz and 20 kHz. In the true magnitude for the right ear, there is a notch at around 15 kHz. There are three models that predict the notch, those models are x_1 , x_4 , and x_5 . In the true magnitude for the left ear, there are no strong notches nor peaks. However, both x_2 and x_5 predicts that there are notches in the frequency span 10 to 20 kHz. Using these magnitudes in a spatial hearing system might lead to misinterpreted positional localization cues.

Moving on to figure 4.8 and starting with the right ear. All predictions are very similar to each other, and it is hard to find any differences as the overall shape is similar and neither has any strong notches or peaks. However, for the left ear there are more differences. The model with the lowest input dimension x_5 , follows the true magnitude surprisingly well and captures the notch at 10 kHz in comparison to the other models that does not capture the notch. The model with input layer x_3 also follows the true magnitude surprisingly well and also captures the two high frequency peaks.

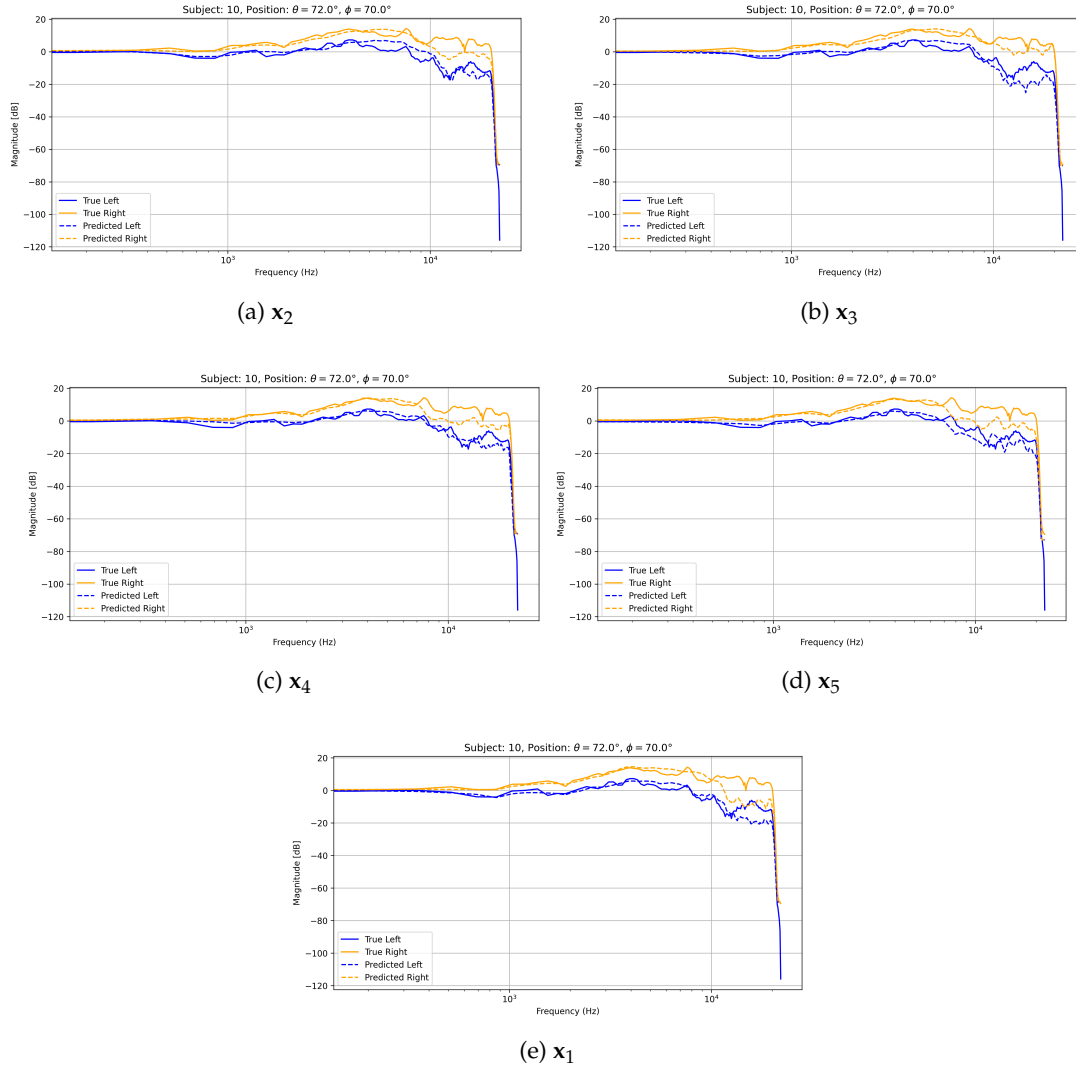


Figure 4.7: True and predicted magnitude for different anthropometric input layers

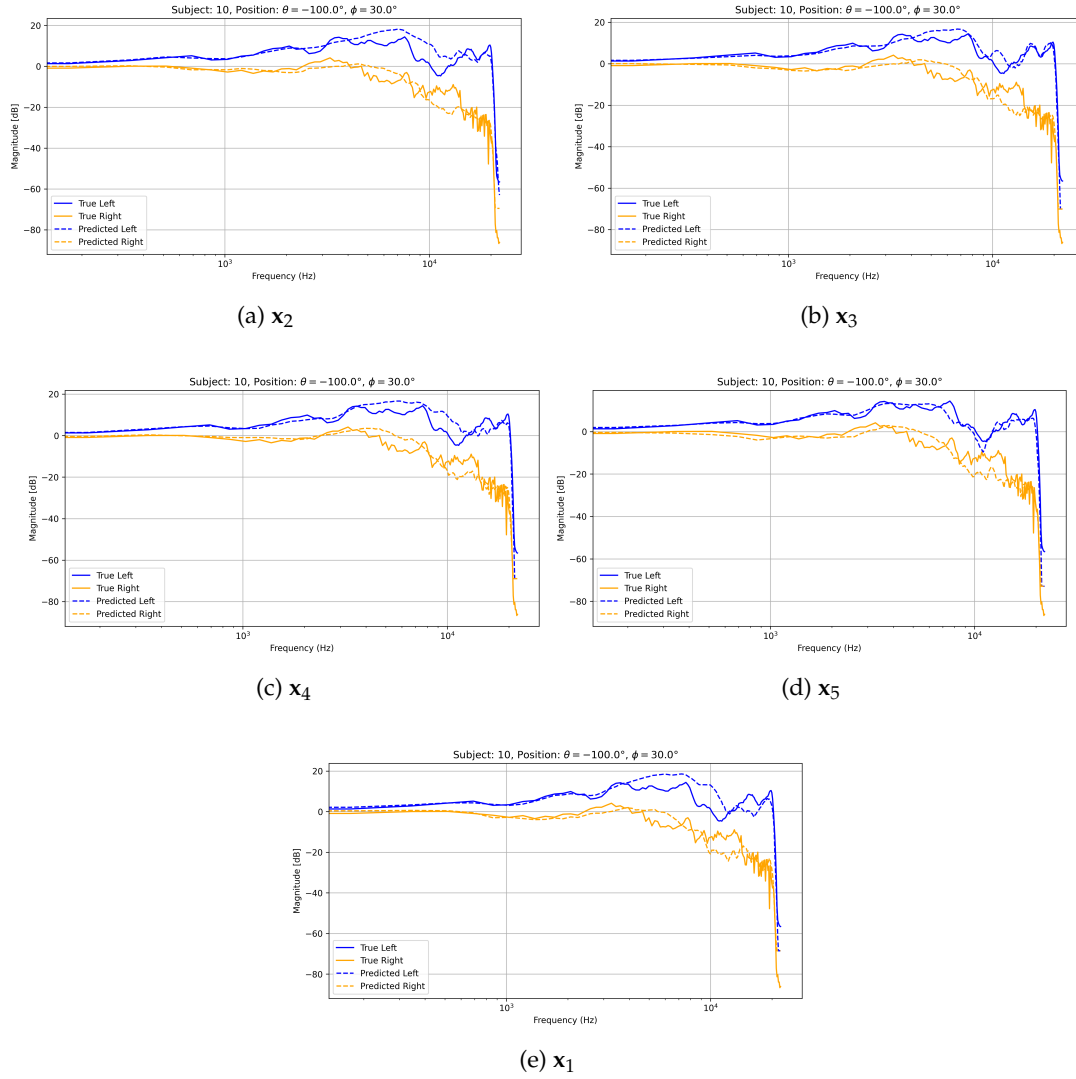


Figure 4.8: True and predicted magnitude for different anthropometric input layers



5 Discussion

This chapter discusses the results from the experiments. The results are compared to each other and the causes of the results are discussed.

5.1 Results

This section analyses and discusses the results obtained for each experiment.

5.1.1 Interpolation Experiment

In the first experiment, the HRTF neural network is the best performing method due to its highly performing spectral accuracy and difference between the ears. Especially for horizontal localization, the HRTF neural network is superior at representing horizontal cues. Furthermore, elevational and antero-posterior cues are also the best performing even though the LSD is not the lowest. Therefore, the recommended method is using the HRTF neural network. Comparing the other three methods is harder. The choice between the three methods is based on what localization cues are prioritized. If the goal only is to interpolate points on the horizontal plane, using bilinear interpolation is recommended as the RMS-ITDE is very low and the RMS-ILDE is reasonable. However, using the neural network might also be a good choice if ILD accuracy is prioritized. If elevational and antero/posterior cues are important, then one might consider using the closest point method.

It is important to note that the experiment is only analytical. Although some magnitudes may differ physically from the true magnitude, it might be perceived completely different by the auditory system.

5.1.2 Personalized HRTF Prediction vs Dataset Selection

From the personalized HRTF prediction vs dataset selection experiment, there were many interesting results.

Starting with the RMS-ITDE for subject 2 which was significantly higher than for the other subjects. The intuition is that the ITD should not be that difficult to predict compared to the magnitude. It gives us a clue that something is wrong with the model. Either, it is the capacity of the model, not enough training data, too few anthropometrics used, or the model finding

the average between subjects. It should not be the capacity of the model as the prediction of the ITD for subject 3 and subject 4 are reasonable and close to the RMS-ITDEs from the interpolation experiment (the lowest RMS-ITDE from the interpolation experiment is $42.1 \mu\text{s}$).

When introducing more data to the model, the results gave that the RMS-ITDE did not decrease with more data. So not enough training data can not be the cause.

This leaves us with two probable causes. The first probable cause is the input layer of the HeadMLP has too few anthropometrics. The second probable cause is that the model learns the average between heads. This is probably due to the dataset being overrepresented near the mean making it difficult for the model to learn a generalization of outliers. To solve this, more anthropometric data can be added from the HUTUBS definition.

So the intuition stated earlier that the ITD is mainly dependant on the head size might be correct, but with the data available, it is not sufficient to accurately predict the ITD for all heads.

Continuing with the magnitude, the average RMS-ILDE for all subjects was higher for the personalized model compared to the mean performing dataset. Similarly to ITD, the ILDE are more dependant on head dimensions rather than ear detail. As both ILDE and ITD lacks accuracy compared to the mean performing dataset. It strengthens the claim that more data anthropometrics is needed for more accurate predictions. Rather than only using head width, height, and depth, a metric for placement of the ears would be interesting to add in the input layer for both the magnitude predictor and ITD predictor. For example, the HUTUBS database contains two metrics for pinna offset labeled x_4 , pinna offset down, and x_5 pinna offset back. These two metrics measure how far back and far down the pinna is located from the middle of the head.

The LSD was lower than the best fitted HRTF dataset for all subjects. However, it is important that the overall shape of the magnitude is similar to the true magnitude to capture notches and peaks important for localization.

To capture the notches better, different measures can be taken. The first measure is to increase the learning rate. However, increasing the learning rate alone leads to a more unstable learning loss curve. The learning rate chosen for this model is carefully chosen to stabilize learning meanwhile not averaging between subjects.

Another measure and something that would be interesting to test is to modify the loss function. The overall structure of the predicted magnitudes follows the true magnitude closely. One could then increase α in the loss function algorithm in algorithm 3.13.

Each subject has 440 points and the training is done on 90 subjects. As each subject has many more points than number of subjects used for training, the model might focus more on accurately predicting the points. From figure 4.5, the trend is that the LSD is decreasing with more training data. It would be very interesting to see how the predicted LSD would change if the number of points was downsampled to be closer to the number of subjects. It would also be interesting to think about adding more subjects for training. It is important that the subjects anthropometric measures closely follows the HUTUBS measurement definitions. Another database that follows these measurement definitions is the SONICOM database [34][35]. However, the database lacks the measurement for the head size x_1 - x_3 . For the SONICOM database subjects to be added to the personalized model, these head measurements need to be measured. The measurements can be done using the 3D scans of the subjects head that is included in the dataset.

5.1.3 Anthropometric Feature Reduction

The anthropometric feature reduction experiment yielded very interesting results. Starting with the LSD for the different models in table 4.3, the models with input layers x_2 , x_3 , and x_4 all have lower LSD than the model using all anthropometrics with layer x_1 . Intuitively, more metrics should give better predictions. However, according to the LSDs, for the chosen model, predictions do not necessarily get better by increasing the anthropometric input layer

dimension. According to the LSDs, one can argue for the opposite, that the model gives worse predictions. On the other hand, decreasing the input dimension more by using x_5 , the LSD increase more. So increasing the input layer size improve predictions to an extent.

5.2 Method

The methods used in this work were chosen to investigate whether different neural-network based interpolation and personalization can improve sound perception. Although the experiments where successful, the methods contain several limitations.

One limitation to this work is ITD extraction using GCC-PHAT. As the time delays between the ears are small, the ITD calculations are sensitive to noise. As a consequence of the sensitivity to the noise, this work becomes highly dependent on the HUTUBS data collection. Furthermore, after removing duplicated subjects and subjects missing anthropometric measurements, the effective subject count is only 90. Also, the HUTUBS HRTF collection was only done on points on the sphere with radius 1.47 m, making the HRTF interpolation lack distance cues. To add distance cues, a distance filter was added with a simplification of the ISO 9613-1:1993 standard was added.

Another limitation is the HRTF reconstruction from the personalized neural network. The method used is constructing a minimum phase system with an added ITD. This method removes all cues related to timing differences due to pinna reflections.

Instead of predicting linear magnitudes, the magnitudes can be preprocessed and converted to decibels. Although lower frequency behaviour might be harder to learn, higher frequency behaviour should in theory be easier to learn. This is due to deviations in the linear magnitude for small amplitudes, translates to large changes in logarithmic domain. This phenomenon is also mirrored in the result as the plots show that the personalized model has problems learning high frequency behaviour of the HRTFs.

5.3 Future Work

Several areas remain for future improvements for the algorithms.

For the personalized neural network, it would be very interesting to add more subjects to training data and investigate if the model improve spectral accuracy.

Another interesting area in the personalized neural network is to investigate how the phase can be better predicted. A proposed solution is to predict how the phase deviates from the linear ITD.

The last improvement for the personalized neural network is to continue to experiment with different anthropometric values, is it possible to add more metrics? Is it possible to add new metrics not defined in the HUTUBS definition?

Another topic worth investigating is what sound sources should be used in a system where different objects needs to be separated. For example, if a visually impaired skier wishes to use a visual aid that plays sound from the border of the course and obstacles. What sounds should be used so the skier can separate sounds based on if it is a course marker or if the sound is an obstacle.

Furthermore, making experiments in controlled environments on humans would add a lot of knowledge about the algorithms and how they actually perform regarding localization accuracy. In this work, only analytical tests and results regarding approximated localization cues have been presented, but not how the sounds are actually perceived by the auditory system.



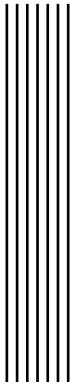
6 Conclusion

The purpose of this thesis was to investigate how *Head-Related Transfer Functions* (HRTFs) could be used in a sonification system. The aim was to compare different interpolation methods for HRTFs, develop a personalized neural network predicting HRTFs based on anatomical features, and determine how the anthropometric input layer could be minimized while maintaining spectral accuracy. The aim of the thesis was achieved through metrics measuring the *Interaural Time Delay Error* (ITDE), *Interaural Level Difference Error* (ILDE), and Log Spectral Distance (LSD). Furthermore, magnitude plots were compared to evaluate how localization cues were suppressed by different methods.

The first research question investigates how different interpolation methods compare when interpolating new HRTFs in a dataset. This thesis showed that neural network predicting the magnitude and the phase separately outperforms all other methods.

The second research question investigates how a personalized neural network improves HRTF dataset selection. The proposed model finds non-linearities between subjects yielding a lower LSD than the most correlating dataset. However, the ITDE and ILDE was on average higher than the average selected dataset. To comfortably say that the personalized model improves HRTF dataset selection, the ITDE and ILDE has to be improved.

The third research question investigates how the anthropometric input layer to the personalized model can be minimized while maintaining acceptable spectral accuracy. This thesis showed that although several metrics were excluded, the LSD remained low and could even decrease while the ILD remained close to equal for all experimental models.



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